

Yoon-Ji Choi

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Research Interests

- Bimanual Robot Manipulation; Vision-Language-Action Models; Multimodal Perception; Humanoid Manipulation

Education

- M.S. in Mechanical Engineering**, Dongguk University, Seoul 2025–Present
Interactive Robotics Lab; Advisor: Prof. Soo-Chul Lim; GPA: 4.5/4.5; expected Aug. 2027.
- B.S. in Mechanical, Robotics and Energy Engineering**, Dongguk University, Seoul 2022–2025
GPA: 3.91/4.5.

Research Experience

- Graduate Researcher**, Interactive Robotics Lab, Dongguk University Sep. 2025–Present
- Developed a bimanual VLA architecture that couples task-aware multi-view selection with interaction-aware action decomposition; evaluated on six RoboTwin 2.0 tasks and three long-horizon real-robot tasks, improving average success by 27.7% in simulation and 43.3% on hardware over a monolithic baseline.
 - Built multimodal robot-learning pipelines using RGB, thermal, tactile, force, and proprioceptive signals; contributed to task-aware sensory routing, thermal-aware VLA, and interaction-force estimation research.
 - Integrated VR-based bimanual teleoperation, RBY1/UR5e control, dexterous-hand retargeting, simulation/real data collection, and VLA fine-tuning/inference pipelines using $\pi 0.5$ and GR00T.

Publications

Peer-Reviewed Journal Article

1. Young-Chae Son, Dae-Kwan Ko, **Yoon-Ji Choi**, and Soo-Chul Lim. 2026. “ThermoAct: Thermal-Aware Vision-Language-Action Models for Robotic Perception and Decision-Making.” *IEEE Robotics and Automation Letters*, 11(5):6106–6113.

Manuscripts Under Review

2. **Yoon-Ji Choi**, Young-Chae Son, and Soo-Chul Lim. 2026. “See Selectively, Act Adaptively: Dual-Level Structural Decomposition for Bimanual Robot Manipulation.” *IEEE Robotics and Automation Letters*, under review. arXiv:2606.13279.
3. Young-Chae Son, Jung-Woo Lee, **Yoon-Ji Choi**, Dae-Kwan Ko, and Soo-Chul Lim. 2026. “Task-Aware Modality Selection for Multimodal VLA in Robotic Manipulation.” *IEEE Robotics and Automation Letters*, under review.
4. Dae-Kwan Ko, Kang-il Yoon, **Yoon-Ji Choi**, Kang-Won Lee, and Soo-Chul Lim. 2026. “Vision-Language-Force Models as Physical Observers: Adaptation for Interaction Force Estimation.” *Information Fusion*, under review.

Conference Presentation

“ThermoAct: Thermal-Aware Vision-Language-Action Models for Robotic Perception and Decision-Making” 2026

Poster presentation, *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026)*, Paper #238.

Selected Research Projects

Data-Efficient Humanoid Autonomous Manipulation Intelligence — National Research Foundation of Korea 2025–Present

- Built a VR-to-humanoid demonstration pipeline for RBY1 and a dexterous hand, including synchronized robot-state and sensor-data collection.
- Fine-tuned GR00T N1.7 with real and simulation-augmented demonstrations and integrated a distributed ROS2 inference/execution pipeline with a VLM planner and sub-task VLA servers.
- Performed real-robot evaluation and trajectory-level failure analysis to connect data composition, hardware behavior, and policy performance.

Language-Conditioned Robot Motion Generation with Visual/Tactile Multimodal Sensing — Ministry of Science and ICT 2025–Present

- Integrated thermal cameras and DIGIT tactile sensors for synchronized RGB, thermal, tactile, force, and robot-state data collection.
- Trained and evaluated multimodal VLA manipulation policies and implemented force-estimation baselines for physical-interaction analysis.

Patent

Co-inventor, “Robot Control Method, Apparatus, System, and Computer-Readable Recording Medium Using Thermal Information” 2026

Korean Patent Application No. 10-2026-0067117; filed Apr. 14, 2026.

Selected Engineering Experience

Bimanual Teleoperation and Streaming IK — UR5e, RBY1, Inspire/RH56F1, Meta Quest, MuJoCo 2026

Integrated dual-arm teleoperation, geometry-based inter-arm collision avoidance, dexterous-hand retargeting, and a state-continuous StreamingIK solver with singularity damping and Cartesian/joint-space motion limits.

Mobile Manipulator Teleoperation and Data Collection System — Scout Mini, ROS, Python, React, WebSocket 2025

Built a distributed remote-control and data-collection framework with real-time camera/sensor streaming and an input-monitoring dashboard for a mobile manipulator.

SOOMAC Robot Arm Team, Founder and Team Lead 2024

Led an eight-member team developing collaborative-robot applications, including image-defined task execution and autonomous hamburger assembly; the team earned four competition awards in 2024.

Research Funding

AI Seoul Tech Research Support Program, Seoul Future Talent Foundation 2026
Selected researcher; one-year research support of KRW 20 million.

Honors and Awards

Chairman Award, Federation of Middle Market Enterprises of Korea (FOMEK), Creative Engineering Design Competition 2025
Best Award, AIoT Creative Engineering Design Competition, Dongguk University

Minister of Trade, Industry and Energy Award, Creative Engineering Design Competition 2024
Best Award, AIoT Creative Engineering Design Competition, Dongguk University
Director General Award (3rd Place), R-BIZ Challenge Zeus Robot Mission
Encouragement Award, Capstone Design Value-Up Program

Innovation Award, Autonomous Driving S/W Advanced Competition, Seoul National University Future Mobility Technology Center 2023
Best Award, AIoT Creative Engineering Design Competition, Dongguk University
Excellence Award, Capstone Design Value-Up Program
Encouragement Award, International Student EV Autonomous Driving Competition (AI Autonomous 1/5 Racer)

Grand Prize, Future Mobility Urban Challenge 2022
Excellence Award, Adventure Design Competition
Excellence Award, IoT Home Appliance Consortium Universities Creative Engineering Design Competition
Award, Creative Engineering Design Competition, Korean Society for Engineering Education

Technical Skills

Programming: Python, C/C++, JavaScript, Node.js

AI / ML: PyTorch, scikit-learn, Vision-Language-Action models, multimodal learning, computer vision

Robotics: ROS / ROS 2, MuJoCo, Isaac Sim, OpenCV, teleoperation, inverse kinematics

VLA / Models: π 0.5, GR00T N1.7

Platforms: UR5e, RBY1, Inspire/RH56F1, Kinova Gen3 Lite, Scout Mini, Meta Quest, Jetson, Raspberry Pi

Languages: Korean (native); English (OPIc Intermediate High, Sep. 2026)